

PIUM – Phase Interference Universe Model

A phase-interference–based model of the universe
A unified wave-field interpretive framework for matter, mass, gravitation, detection, galactic coherence, and quantum correlation

Version 1.3 with Extensions v1.4, v1.5, v1.6 and v1.7
Author: Szabolcs Krehlik
ORCID: 0009-0003-8623-7876
© 2025–2026 Szabolcs Krehlik. All rights reserved.
License: Creative Commons Attribution–NonCommercial–NoDerivatives 4.0 International
CC BY-NC-ND 4.0

ABSTRACT

The Phase Interference Universe Model (PIUM) is a unified ontological interpretive framework in which observable physical phenomena are understood as phase and amplitude configurations of a single universal wave field, Φ(x,t).

In this framework, matter is not treated as a collection of elementary point-like objects. What is conventionally called a particle is interpreted as a stable or quasi-stable closed wave configuration: a localized, phase-coherent structure sustained by internal circulation and global self-consistency. PIUM distinguishes between open wave modes and closed wave modes. Open modes propagate phase and configuration information without forming persistent material structures. Closed modes arise when phase relations allow the field to self-connect into resonant configurations that appear as matter.

Detection is interpreted not as the transformation of a wave into a particle, but as a resonant state change inside the material structure of the measuring apparatus. Measuring devices are themselves composed of closed configurations and therefore participate in the same field ontology as the phenomena they detect.

Mass is interpreted as internal tension within a closed configuration, arising from deviation relative to a phase-neutral null-point state. The null point is not empty space, but a universal reference state and a state of maximal latent potential.

Gravitation is interpreted not as a fundamental force added to matter, but as the macroscopic tendency of closed configurations to relax toward their own phase null-point states. This interpretation does not replace the mathematical structure of General Relativity or modify established gravitational calculations.

The v1.4 extension clarifies PIUM’s compatibility with measurement, gravitational observations, and the Weak Equivalence Principle. The v1.5 extension defines empirical and numerical interfaces, including possible connections to cosmic microwave background structure, gravitational-wave phase coherence, galactic dynamics, and large-scale topology. The v1.6 extension introduces explicit wave-dynamical representations and interprets flat galactic rotation curves as possible signatures of collective phase coherence. The v1.7 extension incorporates quantum entanglement as non-separable phase-correlated closure within the same universal wave-field ontology.

PIUM does not replace the Standard Model, General Relativity, or standard quantum mechanics. It introduces no new particles, no new fundamental forces, no adjustable physical constants, and no alternative predictive equations. Dynamical equations referenced in the extensions are presented only as compatible representations or exploratory numerical interfaces.

The purpose of PIUM is to provide a coherent interpretive layer beneath existing physical formalisms, offering a unified ontological reading of matter, measurement, gravitation, galactic coherence, and quantum correlation.

CORE STRUCTURE

- 1. Foundational assumption
PIUM assumes that the ontological basis of physical reality is a single universal wave field, Φ(x,t), which dynamically interferes with itself.
Particles, fields, and interactions are interpreted as stable or quasi-stable configurations and interaction modes of this field rather than as fundamentally separate entities.
- 2. Open and closed wave modes
Open wave modes are propagating, non-localized configurations of Φ(x,t).
Closed wave modes arise when phase relations permit resonant self-closure, producing localized and persistent structures.
The distinction between open and closed modes is primarily topological and phase-relational rather than merely energetic.
- 3. Detection
Detection is interpreted as a resonant configuration response inside the measuring apparatus.
A detected event is therefore not an ontological conversion of wave into particle, but a local state change in an already material configuration.
- 4. Matter and mass
Matter is phase-coherent closure.
Mass is interpreted as internal tension arising from the deviation of a closed configuration from the null-point reference state.
Quantization is associated with closure conditions and global phase self-consistency.
- 5. Null point
The null point is the phase-neutral reference state of the universal wave field.
It is not equivalent to empty space.
It functions as both a universal rest reference and a state of maximal latent potential.
- 6. Gravitation
Gravitation is interpreted as the macroscopic relaxation tendency of closed configurations toward their null-point states.
This does not alter gravitational dynamics.
It provides an ontological interpretation of gravitational behavior rather than a replacement for General Relativity.
- 7. Structural non-uniformity of space
Because Φ(x,t) interferes with itself, space is structurally non-uniform.
Stable material organization becomes more probable where phase relations support persistent closure.
Large-scale structure may therefore be interpreted as the expression of persistent field-interference organization.
- 8. Matter and antimatter
Matter and antimatter are interpreted as closure configurations sharing the same null-point reference but differing in global phase orientation.
Annihilation corresponds to the cancellation of incompatible closure orientations and return to open wave modes.
- 9. Particles as closure modes
Photon: an open wave mode whose detection corresponds to a state change inside material detector configurations.
Electron: a stable fractional or spiral closure mode associated with spin-1/2 behavior.
Proton and neutron: stable or quasi-stable composite closure configurations whose internal sectors correspond to phase modes rather than separable fundamental constituents.
- 10. Stability and persistence
Stability arises from global phase self-consistency.
Persistence reflects topological closure rather than external confinement.
- 11. Non-reductionism
Physical systems are not reduced to smaller independent entities.
Apparent constituents reflect internal organization modes of a single continuous field.
- 12. Relation to existing theories
PIUM does not contradict the Standard Model, General Relativity, or standard quantum mechanics.
It provides an interpretive layer without altering predictive mathematics.

EXTENSIONS

- v1.4 — Measurement Consistency and Gravitational Validation
The v1.4 extension clarifies that PIUM does not modify established gravitational dynamics. It remains compatible with the Weak Equivalence Principle, classical orbital dynamics, and gravitational-wave observations. Gravitational waves are interpreted as phase-coherence disturbances within the universal wave field, while their observed propagation and waveform properties remain those described by established physics.
- v1.5 — Empirical Interfaces and Validation Pathways
The v1.5 extension identifies possible empirical and numerical interfaces for future testing. These include large-scale CMB phase coherence, gravitational-wave phase structure, galactic dynamics, decoherence signatures in disturbed systems, and topology beyond standard power-spectrum descriptions. The extension does not claim empirical confirmation; it defines where future compatibility, tension, or falsification may occur.
- v1.6 — Explicit Wave Dynamics and Emergent Galactic Coherence
The v1.6 extension introduces a minimal explicit representation of the wave field:
ψ(x,t) = A(x,t) exp(iφ(x,t))
Compatible realizations may include standard wave, Klein–Gordon, Gross–Pitaevskii, or Ginzburg–Landau type forms. These are not introduced as new fundamental laws, but as compatible mathematical settings in which phase coherence, circulation, and closed configurations can be represented.
Closed configurations satisfy a topological phase condition:
∮ ∇φ · dl = 2πn
At galactic scales, PIUM interprets disk galaxies as globally phase-coherent systems in which angular momentum may be redistributed through collective modes. Flat galactic rotation curves are interpreted as possible equilibrium signatures of large-scale phase coherence rather than as modifications of gravitational law.
- v1.7 — Phase-Correlated Closure and Quantum Entanglement
The v1.7 extension interprets quantum entanglement as a non-separable phase-coherent configuration within the universal wave field.
Entangled subsystems are not treated as fully independent closures connected by a later signal. Instead, they are interpreted as components of a single extended configuration subject to global phase constraints.
Observed non-local correlations arise from this non-separability, not from superluminal communication.
Measurement is interpreted as local configuration selection consistent with a globally constrained structure.
PIUM does not modify quantum mechanics, does not introduce hidden signal exchange, and does not claim new empirical predictions. It provides an ontological interpretation compatible with the standard formalism.

CORE STATEMENT

All observable physical phenomena arise from phase and amplitude configurations of a single universal wave field.
Matter is phase-coherent closure, not a collection of fundamentally independent point-like entities. Detection, gravitation, galactic coherence, and quantum correlation are interpreted as different manifestations of the same underlying structural principles.
PIUM provides a unified ontological framework bridging classical, quantum, galactic, and cosmological domains without modifying established physical laws.

END OF SHORT DOCUMENT